



Habib University
shaping futures

Programming Fundamentals

CS 101 L1

Fall Semester 2022

“Everybody should learn how to program because it teaches you how to think” – Steve Jobs¹

“Any fool can write code that a computer can understand. Good programmers write code that humans can understand.” — Martin Fowler²



Course Information

Class Location: GF-E101

Class Meeting Time(s): Tues Thurs (10:30 AM-11:20 AM)

This course fulfills the Habib Liberal Core's "Formal Reasoning" requirement and is part of CS Foundation. It is a required course for all DSSE majors (CE, CS, EE) and for a CS minor. It can also be counted as a free elective.

We are 7 sections in total. Each section meets twice a week for lectures and once a week for a lab. The schedule is as follows.

Lecture	Days	Start	End	Instructor	Room
LEC 1	TR	10:30 AM	11:20 AM	Waqar Saleem	GF-E101
LEC 2	WF	5:30 PM	6:20 PM	Maria Samad	FF-N220
LEC 3	TR	4:30 PM	5:20 PM	Adeel Zafar	GF-E101
LEC 4	TR	10:30 AM	11:20 AM	Neelma Bhatti	E-012
LEC 5	TR	5:30 PM	6:20 PM	Abdullah Zafar	FF-W242
LEC 6	MW	1:30 PM	2:20 PM	Syeda Saleha Raza	SFCL
LEC 7	TR	10:30 AM	11:20 AM	Fahad Javed	FF-W243

Lab	Day	Start	End	Instructor	RA	Room
LAB 1	W	8:30 AM	11:20 AM	Neelma Bhatti	Aisha Batool	E-010
LAB 2	R	8:30 AM	11:20 AM	Maria Samad	Aisha Batool	E-010
LAB 3	F	3:30 PM	6:20 PM	Adeel Zafar	TBD	E-003
LAB 4	W	3:30 PM	6:20 PM	Neelma Bhatti	Faiza Azhar	E-010
LAB 5	W	11:30 AM	2:20 PM	Abdullah Zafar	Faiza Azhar	E-010
LAB 6	R	11:30 AM	2:20 PM	Syeda Saleha Raza	Faiza Azhar	E-003
LAB 7	R	3:30 PM	6:20 PM	Fahad Javed	Faiza Azhar	E-010

Instructor Information



Instructor: Waqar Saleem
Title: Associate Professor, CS
Office Location: C-122
Email: waqar.saleem@sse.habib.edu.pk
Office Hours: TBD

Instructor: Aisha Batool
Title: Research Associate, CS

Office Location: W-300

Email: aisha.batool@sse.habib.edu.pk

Office Hours: Mon, Tue: 12-2

Instructor: Faiza Azhar

Office Location: C-203 pod

Email: faiza.azhar@sse.habib.edu.pk

Office Hours: TBD



Instructor: Adeel Zafar Soomro

Office Location: TBD

Email: adeel.soomro@sse.habib.edu.pk

Office Hours: Tue: 15:30-16:30 and by appointment

Instructor: Fahad Javed

Office Location: TBD

Email: fahad.javed@sse.habib.edu.pk

Office Hours: TBD

Instructor: Maria Samad

Office Location: W-230

Email: maria.samad@sse.habib.edu.pk

Office Hours: Tue/Thur (2:00 pm - 3:00pm)

Instructor: Muhammad Abdullah Zafar

Office Location: TBD

Email: abdullah.zafar@sse.habib.edu.pk

Office Hours: TBD

Instructor: Neelma Bhatti

Office Location: C-217

Email: neelma.bhatti@sse.habib.edu.pk

Office Hours: Tue/Thur (12:00 pm - 1:00pm)

Instructor: Syeda Saleha Raza

Office Location: C-207

Email: saleha.raza@sse.habib.edu.pk

Office Hours: Mon (11:00 am -1:00 pm), Thu (10:30 - 11:30 am), Others: By appointment

Course Description

Access to computing is close to being [a basic human right](#). (also see what [this kid](#) from a decade ago thinks!) Computer Science (CS) studies the various aspects of computing: how computers work, how they can be programmed, how can we make faster computers and write efficient programs, what new problems can we solve with computing, and so on. This first course in CS studies the most exciting aspect: how can we create?

Computers have opened the floodgates for limitless creativity in all endeavors of life: entertainment, science, business, agriculture, finance, education, health, engineering, manufacturing, etc. But in order to make computers do what we want, we have to instruct them in their language¹. There are many such languages, called *programming languages*, and this offering covers [python](#).

Computer scientists like you have made possible things that were seemingly impossible. So while it is important to know your tools, it is equally, if not more, important to be creative and to imagine. This course straddles both extremes and teaches you to bridge the gulf between free thinking creativity and low level, technical details.

Course Aims

This course aims to equip you with the skills to convert your ideas into executable code. Some of these skills are algorithmic thinking and computer programming in a high level programming language. Ultimately, for more interesting and involved ideas, one needs to delve into mathematics, know about data structures, and reason about algorithms, all in order to write more sophisticated code. All of these are covered later in the CS curriculum. But we will see in this course that there is a lot that we can make computers do with basic creativity alone!

Course Learning Outcomes (CLOs)

By the end of the course, students will be able to:

CLO	Outcome	Learning-Domain Level
CLO 01	Design and write programs using the basic programming constructs: interaction, computation, memory, repetition, decision	Cog – 4
CLO 02	Recall the applicable data type to store a given value.	Cog – 1
CLO 03	Make calls to a function, including recursive calls.	Cog – 2

CLO	Outcome	<u>Learning-Domain Level</u>
CLO 04	Use basic data structures: string, list, dictionary, and tuples.	Cog – 3
CLO 05	Fruitfully collaborate on technical problems.	Aff – 3

Mode of Instruction

Instruction in higher education all over the world has vacillated unpredictably between in-person and online for more than two years, leading to at least one important lesson. We do not like remote, online learning or instruction. To every extent possible, this course will take place in person. We will meet twice a week for 50-minute lectures and once a week for a 3-hour lab.

In the unfortunate circumstance where we need to go online, relevant instructions will be shared accordingly. For that contingency, you should have a computer with an internet connection that is capable of running a latest browser version and Zoom.

Required Texts and Materials



Think Python

Edition: 2nd

Authors: Allen B. Downey

Available online at: <https://greenteapress.com/wp/think-python-2e/>

Assessments

A breakdown of course grades (minor changes would be made as the semester progresses):

Assessment type	Quantity	Weight (%)	Remarks
Homework assignments	04	20	The first two assignments are attempted in pairs and the next two, individually. These will be released and submitted on HackerRank.
Lab submission and participation	~15	15	Labs will be attempted in pairs. Labs will typically be on HackerRank.

Assessment type	Quantity	Weight (%)	Remarks
Quizzes	~15	15	There will be an individual quiz in each lab. It will cover the content recently covered in class which, in turn, may rely on previously covered content.
Midterm Exam 1	01	15	The exam will be attempted individually. It will be a programming exam conducted on HackerRank.
Midterm Exam 2	01	15	The exam will be attempted individually. It will be a programming exam conducted on HackerRank.
Final Exam	01	20	The exam will be attempted individually. It will be a programming exam conducted on HackerRank.

You may be called in by the course staff for a viva on your submission for any of the assessments. Your performance on the viva will affect your earned score on the assessment.

Grading Scale

Letter Grade	GPA Points	Percentage
A+	4.00	[95-100]
A	4.00	[90-95)
A-	3.67	[85-90)
B+	3.33	[80-85)
B	3.00	[75-80)
B-	2.67	[70-75)
C+	2.33	[67-70)
C	2.00	[63-67)
C-	1.67	[60-63)
F	0.00	[0, 60]

Note: $[a, b)$ is a range of numbers from a to b where a is included in the range and b is not.

Week-Wise Schedule (Tentative)

The schedule may change in view of class progress as the semester proceeds. All indicated chapters under Reading are from the text book. See the [Live Syllabus](#) for an updated version.

Week	Start	End	Note	Topic	Reading	Lab
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Week	Start	End	Note	Topic	Reading	Lab
1	22 Aug	26 Aug	HW 0 (ungraded)	Introduction; Way of the program	Syllabus; Chapter 1	Hour of Code; Introduction to course tools
2	29 Aug	2 Sep		Variables, expressions, statements; Conditionals	Chapter 2, 5	Practice Problems
3	5 Sep	9 Sep	HW 1 out	Conditionals	Chapter 5	Practice Problems; Quiz: Variables, expressions, statements
4	12 Sep	16 Sep		Functions	Chapter 3	Practice Problems; Quiz: Conditionals
5	19 Sep	23 Sep	HW 1 due	Functions	Chapter 3	Practice Problems; Quiz: Functions
6	26 Sep	30 Sep	Midterm 1; HW 2 out	Functions; Revision	Chapter 6	Midterm 1
7	3 Oct	7 Oct		Recursion	Chapter 5	Practice Problems; Quiz: Functions
8	10 Oct	14 Oct	HW 2 due	Iteration	Chapter 7	Practice Problems; Quiz: Recursion
9	17 Oct	21 Oct	Monday off; HW 3 out	Iteration	Chapter 7	Practice Problems; Quiz: Iteration
10	24 Oct	28 Oct		Strings	Chapter 8	Practice Problems; Quiz: Iteration
11	31 Oct	4 Nov	HW 3 due	Lists	Chapter 10	Practice Problems; Quiz: Strings
12	7 Nov	11 Nov	Midtrerm 2	Lists; Dictionaries	Chapter 10, 11	Practice Problems; Quiz: Lists
13	14 Nov	18 Nov	HW 4 out	Dictionaries	Chapter 11	Practice Problems; Quiz: Lists
14	21 Nov	25 Nov		Tuples	Chapter 12	Practice Problems; Quiz: Dictionaries
15	28 Nov	2 Dec	HW 4 due	Conclusion; Revision		Revision
Finals			Final Exam			

Late Submission Policy

You must submit all assessments no later than the announced deadline. However, with special approval from your instructor, you may submit late within the faculty's assigned deadline with a 20% late submission penalty.

Attendance Policy

You are expected to attend and participate in all lectures and labs. Under extenuating circumstances, you may miss up to 04 synchronous sessions and up to 02 lab sessions. In case of a missed session, you must inform your course instructor and, in case of a lab, also your course RA, with the reason. Failing to do so may raise an early academic alert with the Office of Academic Performance (OAP). Excessive absences will lead to an automatic withdrawal from the course.

Office Hours Policy

Every student enrolled in this course must meet individually with the course instructor during course office hours at least once during the semester. The first meeting should happen within the first five weeks of the semester but must occur before midterms. Any student who does not meet with the instructor may face a grade reduction or other penalties at the discretion of the instructor and will have an academic hold placed by the Registrar's Office.

Academic Integrity

Each student in this course is expected to abide by the Habib University Student Honor Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work.

Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by Habib University regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

- a. Plagiarism: Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. As per University policy, plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated, bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.

- b. Cheating: The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.
- c. Collusion: Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so.

All violations of academic integrity will also be immediately reported to the Student Conduct Office.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

Should copying occur, the student who copied work from another student and the student who gave material to be copied will both be in violation of the Student Code of Conduct.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

Programming is an inherently collaborative activity. This leads to a thin line in this course between academic dishonesty and genuine collaboration. Your instructors will continue to have discussions with you throughout the course on academic honesty and collaboration. Some assessments will be done individually while others will be done in teams. You will be clearly told which is which.

Accommodations for Students with Disabilities

In compliance with the Habib University policy and equal access laws, I am available to discuss appropriate academic accommodations that may be required for student with disabilities. Requests for academic accommodations are to be made during the first two weeks of the semester, except for unusual circumstances, so arrangements can be made. Students are encouraged to register with the Office of Academic Performance to verify their eligibility for appropriate accommodations.

Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. Habib University is committed to providing an atmosphere for learning that respects diversity. While working

together to build this community we ask all members to:

- share their unique experiences, values and beliefs
- be open to the views of others
- honor the uniqueness of their colleagues
- appreciate the opportunity that we have to learn from each other in this community
- value each other's opinions and communicate in a respectful manner
- keep confidential discussions that the community has of a personal (or professional) nature
- use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Habib community

Program Learning Outcomes (For Administrative Review)

Upon graduation, students will have the following abilities:

- PLO 2: Design: Design one or more computer-based solutions of a given problem and select the solution that is best under the circumstances.
- PLO 3: Programming: Program a given solution in a variety of programming languages belonging to different paradigm.
- PLO 4: Implementation: Design and implement software systems of varying complexity.

Program Learning Outcomes (PLOs) mapped to Course Learning Outcomes (CLOs)					
	CLOs of the course are designed to cater following PLOs: PLO 2: Design PLO 3: Programming PLO 4: Implementation PLO 8: Communication and Teamwork				
	Distribution of CLO weightages for each PLO				
	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 2					
PLO 3					
PLO 4					
PLO 8					

Mapping of Assessments to CLOs

Assignments	CLO #01	CLO #02	CLO #03	CLO #04	CLO #05
Homework 01					
Homework 02					
Homework 03					
Homework 04					
Quiz No 01					
Quiz No 02					
Quiz No 03					
Quiz No 04					
Quiz No 05					
Project					
Midterm Exam 01					
Midterm Exam 02					
Final Exam					
Lab					
Student engagement in course activities					

Recording Policy

Only asynchronous and synchronous online sessions will be recorded and uploaded on our Video Management System (Panopto). Link to the folder of recordings will be available to all students. Hyflex classes might be recorded if faculty deems it appropriate.